

HARMAN SINGH MALHOTRA

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SUMMARY

Software Engineer with 1.5+ years designing and shipping production backend microservices and full-stack products in TypeScript, Node.js, and React. Took two greenfield services from empty repository to production ownership, a premium rating engine and a multi-channel communications platform and consolidated member validation across a high-volume registration pipeline. Experienced in distributed systems, RESTful Application Programming Interface (API) design, PostgreSQL, event-driven data pipelines on Google Cloud Platform (GCP), and Continuous Integration / Continuous Deployment (CI/CD). Seeking Software Development Engineer (SDE-1) / full-stack roles.

EDUCATION

Bachelor of Technology in Computer Science (Hons.), AI/ML
UPES . 8.78 CGPA

Aug 2021 – Jun 2025
Dehradun, IN

Indian School Certificate (ISC), Class XII
St. Joseph's Academy . 92%

2021
Dehradun, IN

EXPERIENCE

Software Engineer
Loop Health (Invoq Health)

Dec 2024 – Present
Bengaluru, IN

- Architected Loop's first communications microservice from scratch, dispatching **1M+ emails and WhatsApp messages in under a minute** by streaming job files in 1,000-record batches with back-pressure through parallelised SendGrid and Sinch APIs on Google Cloud Tasks.
- Delivered per-message traceability across **100% of outbound communications**, replacing a fire-and-forget API with a campaign/job/message/event data model and a priority state machine that blocked out-of-order vendor webhooks from corrupting message status.
- Designed a standalone premium Rater microservice on a JSON rules engine with set operators and pro-rata calculators, scaling pricing inputs **from 3 attributes to 18** and replacing per-policy hardcoded functions with operator-uploaded Excel configuration.
- Consolidated two drifting copies of member validation into one three-phase Validation Service serving **40,000–50,000 records/day**, using an adapter layer to onboard the legacy pipeline without a rewrite and Flagsmith flags for zero-downtime rollout.
- Eliminated silent member drop-offs previously caught only through customer escalations by streaming pre- and post-run Google Cloud Pub/Sub events into BigQuery **across 3 services**, adding missed-lives detection queries and integration test coverage.
- Automated Oriental Insurance GMC and GPA endorsement file generation behind a factory pattern with operator-editable configuration, cutting **6 hours of manual preparation per endorsement cycle to near-zero**.

Web Developer Intern
Digital Jalebi

Jun 2024 – Aug 2024
Dehradun, IN

- Reduced database read volume by **40%** on an interactive Event Tech platform by introducing server-side caching and batched queries across a Next.js and Node.js stack driving Internet of Things (IoT) controlled exhibits.
- Improved exhibit load times by **45%** and application stability by **60%** by integrating an on-premise local server that served assets and session state to experiential SaaS displays without an internet dependency.

Project Intern
HERE Technologies

Jun 2024 – Jul 2024
Dehradun, IN

- Engineered a Python and Scrapy crawler spanning 8 live sources, reaching **90% positional accuracy** on location records fed into HERE's mapping data pipeline.
- Increased data refresh frequency by **200%** by building a scheduled scraping framework with normalisation and deduplication, replacing a manual collection process across all 8 tracked sources.

PROJECTS

ExCarte, Natural-language city discovery platform Live . GitHub

- Shipped a full-stack React 18, TypeScript, and Vite application on Supabase/PostgreSQL with Prisma, resolving free-text intent (“cafes in Jayanagar with outdoor seating”) against a curated **1,500+** **venue** Bengaluru dataset.
- Scored per-venue relevance with a self-hosted Gemma 3 large language model (LLM) served over FastAPI, ranking against **60+** structured attributes ingested from Google Places via Scrapy and Selenium.

U-Tour, Autonomous campus tour booking platform Live . GitHub

- Implemented the scheduling gateway for a LIDAR-guided tour robot across a **6-month** campus pilot, using Redis-backed Bull queues to serialise booking requests and merge overlapping tours.
- Prevented double-booking by validating every request against the robot’s point-to-point availability window, and triggered SendGrid status notifications; Node.js, Express, Prisma/PostgreSQL, React, Firebase Auth.

GraphPathGuru, Full-stack graph algorithm visualiser Live . GitHub

- Developed the application end to end on the MERN stack, exposing **14** REST endpoints over Express and MongoDB with JSON Web Token (JWT) authentication, bcrypt hashing, and per-user graph persistence, naming, and sharing.
- Animated algorithm execution step by step through **1,100+** lines of custom frame generators, colouring nodes and edges across visited, comparing, and relaxed states on a drag-and-drop ReactFlow editor.
- Ported **6** shortest-path algorithms (Dijkstra, Bellman-Ford, Floyd-Warshall, Johnson, SPFA, Yen) from C++ subprocesses into a single Rust WebAssembly module via wasm-bindgen, removing the filesystem dependency that had blocked deployment.

SKILLS

Languages	TypeScript, JavaScript, C++, Python, SQL, HTML/CSS
Backend & Frameworks	Node.js, Express.js, React.js, Next.js, FastAPI, Prisma ORM, REST APIs, Microservices, Event-Driven Architecture, Redux, Zustand, TailwindCSS
Databases & Caching	PostgreSQL, Redis, Google BigQuery, Firestore, MongoDB, Supabase, Firebase
Cloud & DevOps	Google Cloud Platform (GCP), Cloud Tasks, Cloud Pub/Sub, Cloud Storage, Secret Manager, Cloud Logging, Docker, Continuous Integration / Continuous Deployment (CI/CD), GitHub Actions, Vercel
Tools & Practices	Git, Jest, Swagger, Feature Flags (Flagsmith), Agile / Scrum, Code Review, Unit & Integration Testing, System Design, Data Structures & Algorithms

CERTIFICATIONS AND ACHIEVEMENTS

Azure AI Engineer Associate, implementing AI solutions using Microsoft Azure	Microsoft
Leetcode, 700+ questions solved at a 1570 rating, proficient in C++	Leetcode
Graph’s Contest Winner, top performer in the university category (UPES)	GraphTheory
Associate Design & VFX Head, guided 9 designers over 20+ projects and 15+ community events	UPES CSI Student Chapter